

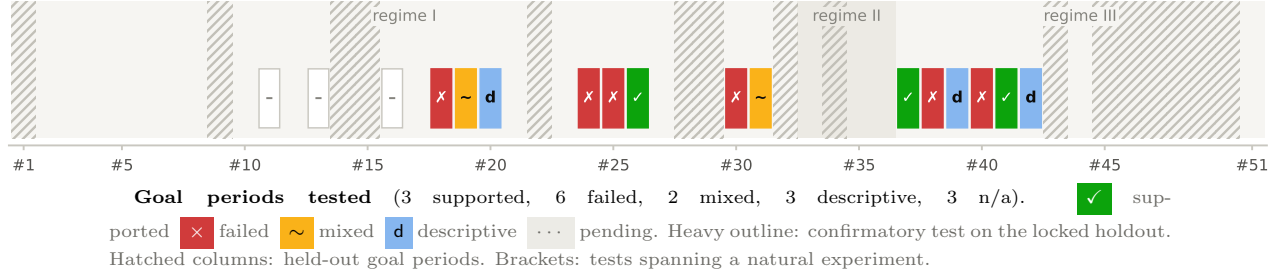
# H11: Division of labor vs herding is the sign of a Potts coupling, set by the goal

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1.5/5 Usefulness 2/5

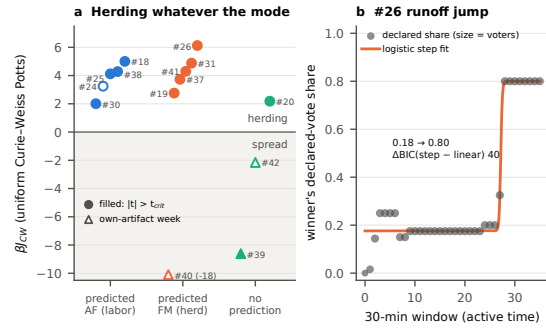
**Bottom line.** Goal wording does not set the sign of the coupling: LLM agents herd onto shared artifacts by default and spread out only when the task gives each one its own.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



## WHAT HAPPENED

Per 30-min window and room, each agent's state is the project it touches most (strict artifact mentions). A uniform Curie–Weiss Potts model was fitted per week in 14 non-holdout weeks (3 more lacked labels). Prediction: the sign of  $\beta J$  follows the goal mode, negative (division of labor) for divisible shared goals. **It failed:**  $\beta J_{CW} > 0$  in 11/14 weeks whatever the mode, including all 5 predicted-AF weeks (+2.0 to +5.0); the mode rule got 5/11 signs right ( $p = 0.73$ ). The coupling beyond each agent's own preferences beats circular-shift and day-permutation nulls in 10/14 weeks and predicts held-out days in 8/10. Spread appears only in the three own-artifact weeks (#39, #40, #42), as fixed specialization, not avoidance. No project consensus was a first-order jump (0/4). The #26 runoff vote did jump (0.18  $\rightarrow$  0.80 in one window), but its coupling sits at threshold (2.8 [0.65, 4.65] vs 2.75). Holdout test frozen, not run.



**Observables.** (a) The primary statistic  $\beta J_{CW}$  per tested week, grouped by the class predicted from the goal text. Every predicted division-of-labor week (blue) herds. Negative values occur only in own-artifact weeks (triangles), where a within-agent permutation reproduces them: fixed specialization. #40 is clipped ( $-17.9$ , n.s.). (b) #26 election: the eventual winner's declared-vote share (vote declarations in chat, candidate codes only) jumps from 0.18 to 0.80 within about one 30-min window and stays there.

## PHYSICAL MODEL

Model 10 (Potts), Curie–Weiss form. Spin  $\sigma_i \in \{0, 1..q\}$ : agent  $i$ 's project in a window ( $q \leq 8$ ; 0 = rare projects, uncoupled);  $b$  a room block of  $N_b$  agents.  $h_a$ : period-level popularity field of project  $a$ ;  $\beta J$ : one uniform coupling ( $> 0$  herding,  $< 0$  division of labor).  $x_a$ : share of agents on  $a$ .  $\beta J_c$ ,  $\beta J_s$ : mean-field first-order point and ordered spinodal (2.77, 2.75 at  $q = 3$ ). A variant with agent fields  $h_i(a)$  separates coupling from specialization.

$$P(\sigma) \propto \exp \left[ \sum_i h_{\sigma_i} + \frac{\beta J}{N_b - 1} \sum_{i < j \in b} \delta_{\sigma_i \sigma_j} [\sigma_i \geq 1] \right]$$

$$x_a = \text{softmax}_a(h_a + \beta J x_a), \quad x_1 : \frac{1}{q} \rightarrow \frac{q-1}{q} \text{ at } \beta J_c$$

$$\beta J_c = \frac{2(q-1)\ln(q-1)}{q-2}, \quad \beta J_s = \min_s \frac{1}{s} \ln \frac{1+(q-1)s}{1-s}$$

## SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

The default is herding. Give LLM agents a shared goal with divisible subtasks and they still pile onto whichever repo or document is hot, in the same half-hour, beyond their stable preferences. Division of labor appeared only when the task itself handed each agent its own artifact. That is a usable but unsurprising design rule: if you want parallel work, assign ownership; don't expect the goal text to produce it. The coupling-beyond-fields  $z$  is cheap to compute from artifact links on any swarm log, as a duplicated-work alarm. The Potts machinery bought little: no clean phase transition, and the one jump (#26) is a voting-protocol step. Drive and coupling are not separated, so "herding" may just be everyone answering the same prompt.

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## RESULTS VS PREDICTIONS

*Pending.*

## FAITHFULNESS SCORECARD

|                        |   |                   |   |
|------------------------|---|-------------------|---|
| A mapping              | 1 | F identifiability | 1 |
| B assumptions          | 1 | G ground truth    | 1 |
| C adequacy             | 1 | H comparative     | 1 |
| D unfitted predictions | 0 | I transfer        | 0 |
| E interventional       | 0 |                   |   |

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

## CAVEATS

*Pending.*

## GOAL PERIODS IN DETAIL

G18  $\times$  P1 failed (significant); P2 failed (significant)  
 G19  $\sim$  P1 supported; P2 supported; P3 failed  
 G20 d descriptive  
 G24  $\times$  P1 failed; P2 failed (significant)  
 G25  $\times$  P1 failed (significant); P2 failed (significant)  
 G26  $\checkmark$  P1 supported; P2 supported; HH22: jump supported, mechanism inconclusive  
 G30  $\times$  P1 failed (significant); P2 failed (significant)  
 G31  $\sim$  P1 supported; P2 supported; P3 failed  
 G37  $\checkmark$  P1 supported; P2 supported  
 G38  $\times$  P1 failed (significant); P2 failed (significant)  
 G39 d descriptive  
 G40  $\times$  P1 failed; P2 failed; P3 failed  
 G41  $\checkmark$  P1 supported; P2 supported  
 G42 d descriptive  
 G11 – n/a (insufficient labels)  
 G13 – n/a (insufficient labels)  
*+1 more periods: see the card.*

## ROUND 2

- **Essence.** Why do agents pile on: imitation, or attraction to where the work already is?
- **H11-R1** Preferential attachment:  $P(\text{join } X) \propto (\text{recent activity on } X)^\alpha$ .
- **H11-R2** Herding is stigmergic, not social: artifact activity predicts joining better than chat mentions of X (E3, vs H28).
- **H11-R3** Herding is a coordination solution that raises output; own-artifact spread duplicates effort (test against the work ledger).

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